

Dominic Pearce

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I am a skilled bioinformatician and software developer with 10+ years of academic and industry experience working at a bioinformatics CRO performing, reporting and communicating analyses for external stakeholders. I headed the R&D team, designing and engineering a large internal codebase to harness R and python to simplify and automate analyses and facilitate the analyst pool's ability to work in a cost-efficient and error-free manner. This work allowed me to both manage and directly contribute to a broad repertoire of projects, developing technical expertise in bioinformatics, biology, -omics technologies, statistics, programming, DevOps, and cloud platforms including AWS, amongst others. I am an excellent written and verbal communicator, with extensive experience in client-facing communication, team leadership and wider strategic technical leadership.

I received my PhD from the University of Edinburgh's Institute of Genetics and Cancer, with a specialty in the applied bioinformatics of cancer. I had the privilege of working closely with clinicians and patients to profile patient-derived tumour material to better understand the mechanism of breast cancer treatment resistance, resulting in several publications in high-impact journals. In addition to my professional career, I've maintained a small but consistent publication record, providing technical expertise for several academic projects.

Experience

Principal Bioinformatics Developer

Fios Genomics

📅 Nov 2022 — February 2026 📍 Edinburgh, UK

- Management of the R&D team.
- Providing technical leadership and directing the R&D strategy.
- Design, development and orchestration of 25+ internal R and python packages, in line with best practices.

Senior Bioinformatics Developer

Fios Genomics

📅 May 2021 — Oct 2022 📍 Edinburgh, UK

- Development of novel internal software in line with best practices.
- Maintenance of 25+ standalone R and python packages.
- Providing technical support to the operations team.

Bioinformatician

Fios Genomics

📅 Nov 2018 — Apr 2020 📍 Edinburgh, UK

- Performing bioinformatic and statistical analysis of various -omics data types.
- Communicating with relevant stakeholders to report and interpret results and define future studies.

References available on request at
dominic.pearce@outlook.com.

Education

University of Edinburgh

📅 Sep 2014 — Jun 2018 📍 Edinburgh, UK

PhD at Institute of Genetics and Cancer. Supervisor: Dr. Andrew Sims. Thesis title: Breast cancer heterogeneity within the neoadjuvant window.

University of Leicester

📅 Sep 2011 — Jun 2012 📍 Leicester, UK

MSc. (distinction) in Cancer Cell and Molecular Biology.

University of Nottingham

📅 Sep 2007 — Jun 2010 📍 Nottingham, UK

BSc. 2.1 in Biochemistry.

Selected publications

- Pearce, D. A. et al. Tumour sampling method can significantly influence gene expression profiles derived from neoadjuvant window studies. *Sci. Rep.*, 2016
- Pearce, D. A. et al. Continuous Biomarker Assessment by Exhaustive Survival Analysis. *bioRxiv*, 2018
- Ross, K. E., Pearce, D. A., McMillan, K. M. What Patterns in Online Classified Puppy Advertisements Can Tell Us about the Current UK Puppy Trade. *Animals*, 2023
- Moir, N, Pearce, D. A, Langdon, S. P & Simpson, I. The significance of molecular heterogeneity in breast cancer batch correction and dataset integration. *Breast Cancer Research*, 2026

Selected projects

Automated reporting infrastructure

To facilitate the reproducible reporting of analyses, I designed and implemented an object-orientated solution to allow data objects (such as the output of a statistical test) to self-report in response to a set of key verbs (*plot*, *table*, *results*, *methods*). In conjunction with a set of curated and parameterised report text templates, this allowed analysts to report results consistently and correctly for free, irrespective of individual experience. These outputs were used to automatically generate Quarto markdown documents, allowing an analysis to seamlessly be reported as a static HTML website, PDF, and/or slide deck.

Reproducible analysis environments

To allow for reproducibility between projects, I implemented a set of Dockerised analysis environments. These formalised and froze all external dependencies necessary for the majority of projects, empowering analysts to begin an analysis immediately and painlessly. Being Dockerised, these environments could in turn be extended by individual analysts, where additional dependencies proved necessary. Importantly, this guaranteed that an analysis could be revisited months or even years later. For full reproducibility, these curated Docker images were hosted via a GitLab container registry, versioned appropriately and tied to the corresponding project repository.

Internal semantic document search

To support cross-team collaboration I designed and built a python package for transformer-based natural language processing (NLP) of internal documents. This was hosted as a Dockerised Shiny app, allowing teams to semantically search legacy proposals, reports and similar documents, making this information easily available for all employees.

Antisense oligonucleotide design pipeline

In collaboration with a large multinational CRO I designed and built an antisense oligonucleotide (ASO) design pipeline, allowing all possible candidate ASOs to be exhaustively assessed *in silico*, in terms of their target binding capacity, non-specific binding potential, etc. This allowed for unsuitable candidate ASOs to be failed early, maximising the cost-efficiency of downstream experimental validation.

Technical Expertise

R	● ● ● ● ●
Python	● ● ● ● ●
Bioinformatics	● ● ● ● ●
Statistics	● ● ● ● ●
ML & AI	● ● ● ● ●

Skills/Exposure

• Team leadership • RNAseq • DNaseq • Proteomics • Metabolomics • Single-cell analysis • Antisense oligonucleotide (ASO) design • Machine learning • AI • LLMs • CDISC standards • Bayesian methods • SQL • LaTeX • Markdown

Methodology/Approach

• Agile • Test-driven development • Literate programming • Reproducible environments • Scalable workflows & automation

Tools

• Linux • Git • GitLab • GitHub • Quarto • Shiny • Vim • tidyverse • tidymodels • Bioconductor • Docker • Nextflow • Snakemake • MySQL • DuckDB • Poetry • uv • scikit-learn • transformers • pandas • numpy

Interests

• Over-the-board chess
• 5-a-side football
• My dogs, cat and children